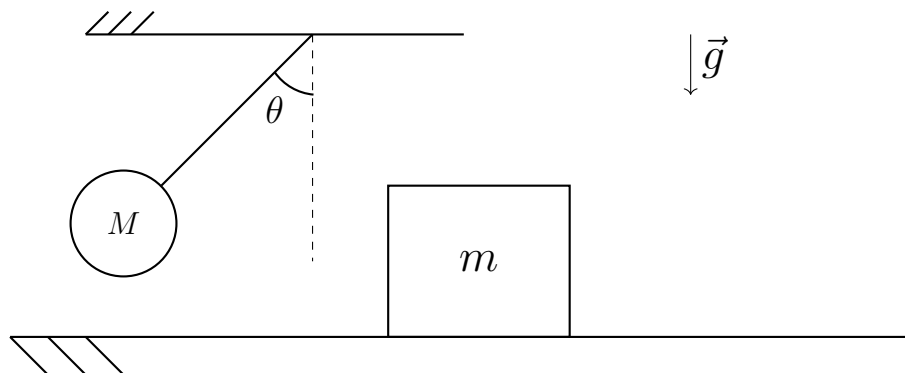


Pendulum with block

A simple pendulum of mass M and length L is released at an angular amplitude θ . Upon reaching the lowest position of its trajectory, it collides elastically with a block of mass m and remains static in its equilibrium position. The block stops after traveling a distance d due to friction with the surface of dynamic friction coefficient μ . Determine the distance traveled by the block.



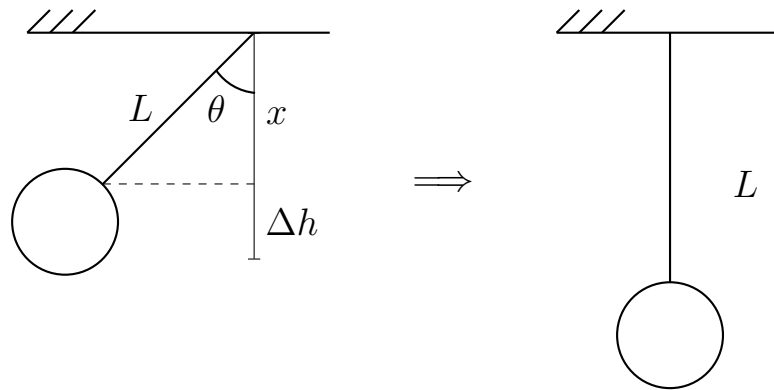
Considering a closed system, the energy is conserved. The energy of the pendulum - initially, just gravitational potential - was partially transformed in kinetic, until the collision with the block, in the lowest point of trajectory. On this point, if the pendulum remains static after the collision, the total kinetic energy was transferred to the block. Thus, the energy absorbed by the block is equal to the variation of the gravitational potential of the pendulum, which is proportional to your variation in height. So, taking U_o as the initial gravitational potential energy of the pendulum, K_c as the kinetic energy of the pendulum in the moment of the collision and U_f as the final gravitational potential energy of the pendulum, we have

$$U_o = K_c + U_f \Rightarrow K_c = U_o - U_f = -\Delta U = Mg\Delta h$$

$$K_c = Mg\Delta h$$

With Δh being the variation of the height of pendulum.

Pendulum geometry



From the picture, it is possible to see that $x = L \cos \theta$ and $L = x + \Delta h$, so

$$\Delta h = L - x = L - L \cos \theta$$

$$\Delta h = L(1 - \cos \theta)$$

Now, we know that the value of the kinetic energy transferred to the block

$$K_c = MgL(1 - \cos \theta)$$

Finally, knowing that the block travels a distance d under the action of the friction force, what is represented by the expression $F_f = \mu \cdot N$, where N means the normal force, imposed by the floor under the block, we can set the energy transferred to the block and the work by the friction force (W_f) equals to each other. So that

$$|W_f| = K_c \Rightarrow F_f \cdot d = MgL(1 - \cos \theta)$$

The normal force is the weight of the block, what is equal to $m \cdot g$

$$\mu mgd = MgL(1 - \cos \theta)$$

Dividing both sides by μmg , we describe the distance traveled by the block.

$$d = \frac{ML(1 - \cos \theta)}{\mu m}$$